

AI/ML Data Annotation Intern – Question 2

Pixel-Level Annotation Methodology

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Objective: The objective was to produce accurate pixel-level masks for compact astronomical sources (blobs/stars) and elongated streak-like space objects from the provided FITS imagery, and to export the resulting annotations in YOLO Ultralytics segmentation format.

1. Methodology Overview

The provided astronomical imagery consisted of FITS images with an original size of 9568×6380 pixels. A programmatic preprocessing pipeline was developed to read the FITS data, estimate and remove the local background, generate candidate detections, divide the imagery into exact 1024×1024 tiles, and support manual pixel-level annotation.

For the final YOLO Ultralytics segmentation dataset, the object classes were:

- **Class 0 – Blob / Star**
- **Class 1 – Streak / Space Object**

Background was represented by the absence of a segmentation polygon rather than as a separate YOLO object class. The automated detection stage was used as a candidate-generation aid. Final class assignment and mask refinement were performed through visual inspection and manual pixel-level annotation.

2(a). Image Pre-processing

FITS Reading and Validation

Each FITS image was read using `astropy.io.fits` with `memmap=False` and converted to `float32`. A dimensionality check was applied to ensure that the input was a two-dimensional image.

The supplied imagery had dimensions:

Parameter	Value
Image width	9568 pixels
Image height	6380 pixels
Data type used for processing	float32
Image dimensionality	2-D

Local Background Estimation

Astronomical imagery contains a spatially varying background and detector noise. A Gaussian filter with $\sigma = 25$ was therefore applied to estimate the smooth local background:

$$B(x, y) = G_{\sigma=25} * I(x, y)$$

The background-subtracted residual was calculated as:

$$R(x, y) = I(x, y) - B(x, y)$$

This enhanced compact sources and elongated features relative to the slowly varying background.

Robust Noise Estimation

The noise level was estimated using the Median Absolute Deviation (MAD):

$$\sigma_{\text{robust}} = 1.4826 \times \text{median}(|R - \text{median}(R)|)$$

If the robust estimate was zero or non-positive, the standard deviation of the residual image was used as a fallback.

Candidate Detection

A high-confidence candidate threshold was calculated as:

$$T = \text{median}(R) + 15\sigma_{\text{robust}}$$

Pixels satisfying

$$R(x, y) > T$$

were converted into a binary candidate image.

Eight-connected component analysis was then applied. Components smaller than 3 pixels were excluded from the candidate set.

The resulting candidates were used only as a **preliminary detection aid**; they were not treated as the final ground-truth annotations.

Visual Enhancement

The annotation interface provided multiple visualization modes:

- Raw image
- Enhanced contrast image
- Background-subtracted image

Percentile-based normalization was used for enhanced visualization so that faint sources could be inspected more easily.

2(b). Handling Image Boundaries and Tiling

The original dimensions were:

$$9568 \times 6380$$

Since these dimensions are not exact multiples of 1024, directly cropping at 1024-pixel intervals would discard pixels along the right and bottom boundaries.

The required tile dimensions were calculated using the ceiling operation:

$$N_{\text{rows}} = \left\lceil \frac{6380}{1024} \right\rceil = 7$$

$$N_{\text{columns}} = \left\lceil \frac{9568}{1024} \right\rceil = 10$$

Therefore, each original image produced:

$$7 \times 10 = 70$$

tiles.

For the ten supplied images:

$$10 \times 70 = 700$$

tiles were generated.

Padding Strategy

The required padded dimensions were:

$$H_p = 7 \times 1024 = 7168$$

$$W_p = 10 \times 1024 = 10240$$

Padding was added only to the bottom and right boundaries. Edge-value padding was used rather than zero padding so that artificial black borders were not introduced into the astronomical background.

Each tile was then extracted using:

$$x_0 = c \times 1024$$

$$y_0 = r \times 1024$$

and

$$Tile = I[y_0 : y_0 + 1024, x_0 : x_0 + 1024]$$

Every generated tile was checked to ensure a final size of exactly 1024×1024 pixels.

Reconstruction Verification

Tile position information was retained so that the tiled imagery could be reconstructed. The completed repatching process reconstructed all ten source images to their original dimensions of:

$$9568 \times 6380$$

The repatching output therefore provided a visual check that the tiling process preserved the complete spatial extent of the original images.

2(c). Distinguishing Streaks from Blobs

Connected components were characterized using their bounding-box dimensions. A preliminary aspect ratio was calculated as:

$$AR = \frac{\max(W, H)}{\min(W, H)}$$

A component was preliminarily suggested as a streak when:

$$AR \geq 3$$

and its major dimension was at least approximately 8 pixels.

Components not satisfying these conditions were preliminarily suggested as blobs.

However, this geometric rule was **not used as the final classifier**. This was necessary because short or partially visible streaks can have a relatively low aspect ratio.

Final Practical Criteria

Blob / Star:

- Compact, approximately point-like appearance.
- Concentrated light source.
- Relatively small spatial extent.
- No clear elongated trajectory.

Streak / Space Object:

- Clearly elongated morphology.
- Greater extent in one direction.
- Linear or track-like appearance.
- The complete visible feature was painted as a pixel-level mask.

Short/fat streaks were therefore handled using visual morphology rather than relying exclusively on the aspect-ratio threshold.

2(d). Criteria for Faint and Small Blobs

The 15σ threshold and 3-pixel component-size criterion were used for candidate generation and were **not hard exclusion rules** for manual annotation.

A faint or small source was labelled as a Blob/Star when:

1. It appeared as a compact, localized feature;
2. It could be visually distinguished from the surrounding background or noise;
3. Its morphology was consistent with a point-like source rather than a streak;
4. Its visible pixels could be localized sufficiently to create a meaningful pixel-level mask.

The annotation interface allowed the brush size to be reduced to approximately one pixel for small features.

An important assumption was:

A faint feature was annotated as a star/blob only when it could be visually distinguished from random background fluctuations. A feature was not treated as a confirmed astronomical object solely because it exceeded an automated intensity threshold.

3. Pixel-Level Annotation Implementation

A custom annotation interface was used rather than relying on rectangular bounding boxes. The annotator could:

- Paint Blob/Star pixels;
- Paint Streak/Space Object pixels;
- Erase incorrect pixels;
- Undo and redo annotations;
- Zoom and pan the image;
- Switch between raw, enhanced and background-subtracted views;
- Display candidate detections as optional visual guidance.

The internal pixel mask used three discrete values:

$$0 = \textit{Background}, \quad 1 = \textit{Blob}, \quad 2 = \textit{Streak}.$$

During the final YOLO conversion, background pixels were excluded and the object classes were mapped to the two-class YOLO scheme:

$$0 = \textit{Blob}, \quad 1 = \textit{Streak}.$$

A technical display issue occurred because the browser annotation canvas could use a smaller display resolution than the source tile. The annotation mask was therefore mapped back to the original 1024×1024 tile using nearest neighbour interpolation so that class labels remained discrete and spatially aligned.

4. YOLO Segmentation Output and Validation

The completed masks were converted to YOLO Ultralytics segmentation format.

The validation stage reported:

Metric	Result
Total images	700
Label files	391
Annotated images	390
Empty/background images	1
Total annotations	8,144
Blob annotations	8,088
Streak annotations	56
Invalid label lines	0
Invalid coordinates	0
Orphan labels	0

The YOLO label syntax and normalized coordinates were successfully validated. The dataset-level validation also identified 309 image tiles without label files; these correspond to tiles that were not manually annotated within the available annotation period.

Annotation Completion

The complete dataset was tiled into 700 1024×1024 tiles. Due to the available annotation time, 390 tiles containing manually verified annotations were completed. One additional tile had an empty label file and therefore contained no positive YOLO objects. The final YOLO export contains 8,144 segmentation instances: 8,088 blob instances and 56 streak instances. The remaining 309 image tiles were retained as image data but were not manually annotated and therefore do not have corresponding YOLO label files.

This limitation is explicitly reported rather than treating unannotated tiles as manually verified ground truth.

5. Assumptions and Limitations

- The FITS images were treated as two-dimensional grayscale astronomical imagery.
- The automated thresholding stage was considered a candidate-generation method rather than a ground-truth classifier.
- Visual morphology was given priority when distinguishing short/fat streaks from compact blobs.
- Faint features were annotated only when they could be reasonably distinguished from background fluctuations.
- Edge padding was used only for tile generation and was not considered part of the original scientific image extent.
- Only manually verified annotations were included as positive segmentation instances.

6. Submission Contents

The accompanying ZIP package is organized as follows:

```
DIGANTARA_ABHINAND_ANNOTATION_SUBMISSION/
|
|-- 01_REPORT/
|   '-- Digantara_Question_2_Report.pdf
|
|-- 02_YOLO_DATASET/
|   |-- images/
|   |-- labels/
|   '-- data.yaml
|
|-- 03_REPATCHED_FULL_SIZE/
|   '-- 10 reconstructed PNG images
|
|-- 04_VALIDATION/
|   |-- annotation_statistics.csv
|   '-- dataset_validation_report.txt
|
|-- 05_SOURCE_CODE/
|   |-- preprocessing_and_detection.py
|   |-- pixel_annotation_app.py
|   |-- qa_annotations.py
|   |-- convert_masks_to_yolo.py
|   |-- convert_npy_images_to_png.py
|   |-- qa_yolo_dataset.py
|   |-- final_yolo_visual_qa.py
|   |-- repatch_full_size.py
|   |-- validate_final_dataset.py
|   '-- requirements.txt
|
'-- README.txt
```

The source code and supporting files are included separately from the written report so that the written response remains within the requested four-page limit.

7. Conclusion

The implemented workflow combines programmatic astronomical-image preprocessing with manual pixel-level verification. Local background subtraction, robust noise estimation and connected-component analysis were used to identify candidate features, while visual inspection was retained as the final decision mechanism for distinguishing stars from streak-like objects.

The complete 9568×6380 imagery was successfully divided into 700 exact 1024×1024 tiles and subsequently reconstructed into ten full-size images. Within the available annotation period, 390 tiles were manually annotated and one additional label file was empty. The final YOLO export contains 8,144 segmentation instances, including 8,088 blobs and 56 streaks, using YOLO class 0 for Blob/Star and class 1 for Streak/Space Object.